

Exercises (with Solutions) · Chapter 2

1. Problem

Consider the following linear regression

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_p X_{ip} + \epsilon_i,$$

where $X_{i1} = 1$ for all $i = 1, \dots, n$.

- (a) Show that Assumption 2 (exogeneity) is violated if some of the regressors are correlated (i.e. $\text{Cov}(X_{ik}, \epsilon_i) \neq 0$ for at least one $k = 0, \dots, p$) with the error term.

Solution

- (a) The exogeneity assumption states that

$$E(\epsilon_i | X_i) = 0$$

implies that

$$\text{Cov}(\epsilon_i, X_{ik}) = 0 \quad \text{for all } k = 0, \dots, p,$$

since

$$\begin{aligned} \text{Cov}(\epsilon_i, X_{ik}) &= E(\epsilon_i X_{ik}) - \overbrace{E(\epsilon_i)}^{=0} E(X_{ik}) \\ &= E(\epsilon_i X_{ik}) \\ &= E(E(\epsilon_i X_{ik} | X_i)) \\ &= E(X_{ik} \underbrace{E(\epsilon_i | X_i)}_{=0}) = 0 \quad \text{for all } k = 0, \dots, p. \end{aligned}$$

Thus

$$\underbrace{E(\epsilon_i | X_i) = 0}_A \Rightarrow \underbrace{\text{Cov}(\epsilon_i, X_{ik}) = 0 \quad \text{for all } k = 0, \dots, p}_B$$

Remember: Any logical implication (A implies B)

$$A \Rightarrow B$$

has a logically equivalent contrapositive (not B implies not A):

$$\neg B \Rightarrow \neg A$$

Therefore, we also have that

$$\underbrace{\text{Cov}(\epsilon_i, X_{ik}) \neq 0 \quad \text{for at least one } k = 0, \dots, p}_{\neg B} \Rightarrow \underbrace{E(\epsilon_i | X_i) \neq 0}_{\neg A}$$

I.e., if any of the regressors X_{ik} , $k = 0, \dots, p$, correlates with the error term ϵ_i , then the exogeneity assumption is violated.

2. Problem

- (a) Prove that

$$\bar{X} = \arg \min_{a \in \mathbb{R}} \sum_{i=1}^n (X_i - a)^2$$

where $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ is the sample mean of the iid random sample $\{X_1, \dots, X_n\}$.

(b) Prove that

$$\sum_{i=1}^n (X_i - \bar{X})^2 \leq \sum_{i=1}^n (X_i - \mu)^2,$$

where $\mu = E(X_i)$ is the population mean of $X_1, \dots, X_n$.

Solution

(a) Let

$$\text{RSS}(a) = \sum_{i=1}^n (X_i - a)^2.$$

Taking the derivative with respect to a yields:

$$\begin{aligned} \text{RSS}'(a) &= \sum_{i=1}^n 2(X_i - a)(-1) \\ &= -2 \sum_{i=1}^n X_i + \sum_{i=1}^n 2a \\ &= -2 \sum_{i=1}^n X_i + 2na \end{aligned}$$

Setting $\text{RSS}'(a)$ equal to zero determines the minimizing argument, say $\tilde{a}$, of $\text{RSS}(a)$:

$$\begin{aligned} \text{RSS}'(\tilde{a}) &\stackrel{!}{=} 0 \\ -2 \sum_{i=1}^n X_i + 2n\tilde{a} &\stackrel{!}{=} 0 \end{aligned}$$

Solving for $\tilde{a}$ yields:

$$\begin{aligned} -2 \sum_{i=1}^n X_i + 2n\tilde{a} &= 0 \\ n\tilde{a} &= \sum_{i=1}^n X_i \\ \tilde{a} &= \frac{1}{n} \sum_{i=1}^n X_i = \bar{X} \end{aligned}$$

Thus, the minimizing argument of $\text{RSS}(a) = \sum_{i=1}^n (X_i - a)^2$ is $\tilde{a} = \bar{X}$.

(b) Since $\bar{X} = \arg \min_{a \in \mathbb{R}} \sum_{i=1}^n (X_i - a)^2$, it follows that

$$\sum_{i=1}^n (X_i - \bar{X})^2 \leq \sum_{i=1}^n (X_i - \mu)^2,$$

where $\mu = E(X_i)$ is the population mean of $X_1, \dots, X_n$.

3. Problem

Suppose we have a data set with dependent variable

$Y = \text{“starting salary after graduation”}$

and four predictors

- $X_1 = \text{GPA}$
- $X_2 = \text{IQ}$
- $X_3 = \text{Gender}$ (1 for Female and 0 for Male)
- $X_4 = X_1 X_2$ Interaction between GPA and IQ
- $X_5 = X_1 X_3$ Interaction between GPA and Gender

Suppose we use least squares to fit the model and get the following estimates:

$$\hat{\beta}_0 = 12, \hat{\beta}_1 = 24, \hat{\beta}_2 = 0.04, \hat{\beta}_3 = 20, \hat{\beta}_4 = 0.04, \text{ and } \hat{\beta}_5 = -8.$$

Thus we have:

$$\hat{E}[Y|X] = \hat{f}(X) = 12 + 24 \overbrace{\text{GPA}}^{X_1} + 0.04 \overbrace{\text{IQ}}^{X_2} + 20 \overbrace{\text{Gender}}^{X_3} + 0.04 \overbrace{\text{GPA} \cdot \text{IQ}}^{X_4 = X_1 \cdot X_2} - 8 \overbrace{\text{GPA} \cdot \text{Gender}}^{X_5 = X_1 \cdot X_3}$$

Show that, for a fixed value of IQ and GPA, males earn more on average than females provided GPA is greater than 2.50

Solution

(a)

$$\begin{aligned} & \text{(male)} > \text{(female)} \\ & 12 + (24)X_1 + (0.04)X_2 + (20) \cdot 0 + (0.04)X_4 + (-8) \cdot 0 \cdot X_1 > 12 + (24)X_1 + (0.04)X_2 + (20) \cdot 1 + (0.04)X_4 + (-8) \cdot 1 \cdot X_1 \\ & 12 + (24)X_1 + (0.04)X_2 + 0 + (0.04)X_4 + 0 > 12 + (24)X_1 + (0.04)X_2 + (20) + (0.04)X_4 + (-8)X_1 \\ & 12 + (24)X_1 + (0.04)X_2 + (0.04)X_4 > 32 + (16)X_1 + (0.04)X_2 + (0.04)X_4 \\ & (8)X_1 > 20 \\ & X_1 > 2.5 \end{aligned}$$

4. Problem

(a) Consider the linear regression model $Y = X\beta + \epsilon$. Let

$$Y = \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} \quad \text{and let} \quad M_X = \begin{pmatrix} 0.5 & 0.0 & -0.5 \\ 0.0 & 0.0 & 0.0 \\ -0.5 & 0.0 & 0.5 \end{pmatrix}$$

What is the value of the residual sum of squares $\sum_{i=1}^3 e_i^2$?

Solution

(a)

$$\begin{aligned} e &= M_X Y \\ &= \begin{pmatrix} 0.5 & 0.0 & -0.5 \\ 0.0 & 0.0 & 0.0 \\ -0.5 & 0.0 & 0.5 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} = \begin{pmatrix} -0.5 \\ 0 \\ 0.5 \end{pmatrix} \end{aligned}$$

Thus,

$$\sum_{i=1}^3 e_i^2 = 0.25 + 0 + 0.25 = 0.5.$$